QFRM Case 1

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1 Data handling and descriptive statistics

We construct a daily portfolio with exposure to equity, FX, and interest-rate risk. The dataset contains three stocks (IBM, MSFT, MTOR), the S&P 500, the EUR/USD exchange rate, and a loan proxy based on the 3-month Treasury bill rate. To satisfy the assignment requirement of a synchronized portfolio sample, only dates with complete information across all series are retained.

The stock and index data were already provided as daily total return series and were used directly after standardizing dates. The EUR/USD series was provided in levels and converted into daily simple returns as

$$r_t^{FX} = \frac{FX_t}{FX_{t-1}} - 1.$$

Isolated missing observations in level-type series were filled only when a missing value was surrounded by two valid observations, in which case the local average was used. This limited interpolation was applied conservatively and only where economically defensible.

The loan component was modeled as a simple floating-rate proxy linked to the short rate. After converting the 3-month Treasury bill yield into decimal form, daily loan returns were computed as

$$r_t^{loan} = \frac{y_t + s}{252},$$

where y_t is the observed Treasury yield and $s = 0.015$ is a fixed credit spread of 150 basis points.

All series were then merged by date. After merging, all dates with remaining missing values were dropped to preserve a fully synchronized sample for joint portfolio risk measurement. The final sample contains 2,449 daily observations from 30 March 2011 to 31 December 2020.

Table 1 reports descriptive statistics for the synchronized daily returns in percentage points. The loan proxy is, as expected, the least volatile series. Among the risky assets, MTOR is the most volatile, followed by MSFT and IBM, while the S&P 500 is less volatile than the individual stocks. EUR/USD is less volatile than equities but clearly more variable than the loan component. Kurtosis is elevated for all risky assets, indicating fat tails and providing an early signal that non-Gaussian risk models may be relevant in later VaR and ES analysis.

Table 1: Univariate descriptive statistics of synchronized daily risk-factor returns

Series	Mean (%)	SD (%)	Min (%)	P5 (%)	Median (%)	P95 (%)	Max (%)	Skew.	Kurt.
EUR/USD	-0.004	0.514	-2.131	-0.817	0.000	0.829	3.112	0.004	5.351
Loan	0.008	0.003	0.006	0.006	0.008	0.013	0.016	1.152	2.754
IBM	0.013	1.439	-12.016	-2.262	0.035	2.203	11.301	-0.428	12.890
MSFT	0.110	1.623	-14.778	-2.298	0.113	2.531	14.217	0.016	13.371
MTOR	0.065	3.216	-23.753	-4.856	0.000	4.878	22.105	0.320	8.375
S&P 500	0.056	1.098	-9.691	-1.496	0.066	1.566	9.321	-0.656	19.712

Notes: Statistics are computed on the synchronized daily sample from 30 March 2011 to 31 December 2020. Mean, standard deviation, extrema, median, and quantiles are reported in percentage points. Skewness and kurtosis are unit-free.

2 Portfolio Weights

The weights of the portfolios stay the same during the sample period. The equity block makes up half of the portfolio, with 20% in IBM, 20% in MSFT, and 10% in MTOR. The smaller amount given to MTOR is because its returns are far more volatile than those of the other two equities, as seen in Table 1. The S&P 500 gets 20% of the weight, which gives it a wide range of exposure to the market. The other 30% is split between EUR/USD (15%) and the loan component (15%), ensuring the portfolio has exposure to both currency and interest rate

risk as well. We choose fixed weights on purpose. Allowing for rebalancing would change the returns in ways that make it harder to figure out who is responsible for the risk and make it harder to understand VaR and ES estimations throughout the whole estimation frame. This makes the risk analysis clear: the same positions are held the whole time, so any changes in VaR and ES over time are due to changes in the market, not in the portfolio itself.

3 VaR and ES Methods

We estimate one-day VaR and ES at the 99% confidence level using five methods. All methods are applied to the portfolio return series directly, with losses defined as $L_t = -r_t^p$.

3.1 Normal Variance-Covariance

We start with the normal variance-covariance method as a benchmark, eventhough the kurtosis figures in Table 1 already suggest normality is unlikely for this portfolio. Under normality, VaR and ES follow from the portfolio mean and standard deviation:

$$\mathrm{VaR}_\alpha = -\mu + z_\alpha \sigma, \qquad \mathrm{ES}_\alpha = -\mu + \sigma \frac{\phi(z_\alpha)}{1-\alpha},$$

where z_α is the α-quantile of the standard normal and $\phi(\cdot)$ its density. This gives VaR = 2.13% and ES = 2.45%. We treat these as a lower bound baseline; given the excess kurtosis documented in Table 1, the true risk is likely higher.

3.2 Student-t Variance-Covariance

Given the excess kurtosis across all series, replacing the normal with a Student-t distribution could be a better assumption. We estimate VaR and ES for $\nu \in \{3, 4, 5, 6\}$ degrees of freedom and adjust the scale parameter as $c = \sigma/\sqrt{\nu/(\nu-2)}$ to keep the variance comparable. QQ-plots for each specification show that $\nu = 5$ fits the empirical loss distribution most closely in both tails, $\nu = 3$ is too heavy, $\nu = 6$ underestimates the extremes, so we use $\nu = 5$ going forward.

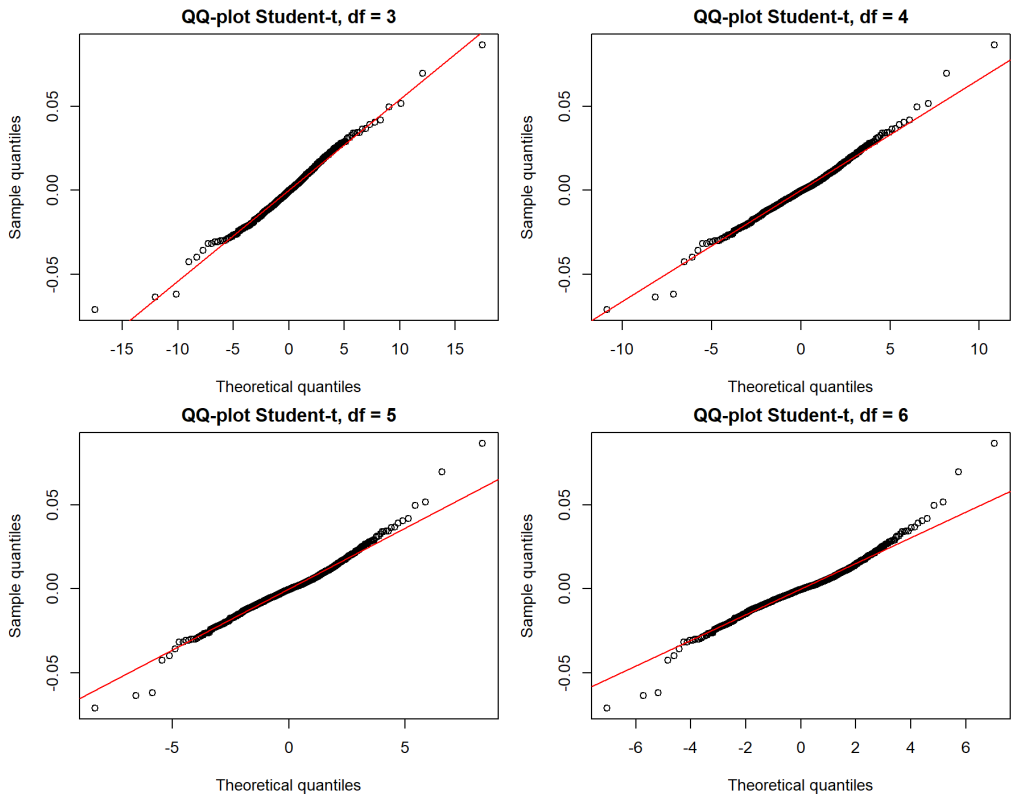

Figure 1: QQ-plots of portfolio losses against Student-t distributions with $\nu \in \{3, 4, 5, 6\}$ degrees of freedom. The red line indicates the theoretical quantiles scaled by the estimated scale parameter. $\nu = 5$ provides the closest fit in the tails.

3.3 Historical Simulation

The historical simulation makes no distributional assumptions at all. VaR is just α-quantile of observed losses, and ES as the average of everything beyond that threshold:

$$\widehat{\mathrm{VaR}}_\alpha = L_{(k)}, \qquad \widehat{\mathrm{ES}}_\alpha = \frac{1}{T-k+1} \sum_{t=k}^{T} L_{(t)},$$

where $k = \lfloor \alpha T \rfloor$ and losses are sorted from smallest to largest. There are no distribution needs and no parameters estimated. The limitation is that it is entirely backward-looking. In our case, the COVID crash of march 2020 sits in the tail of the sample, which is why historical simulation produces the highest VaR and ES of the 5 methods.

3.4 GARCH(1,1) with Constant Conditional Correlation

The previous three methods treat volatility as constant, which might not always be true if you're looking at financial return series. The GARCH(1,1) + CCC model addresses this by estimating a separate GARCH(1,1) process for each asset, giving time-varying conditional volatilities $\sigma_{i,t}$. We then follow the CCC approach of McNeil, Frey and Embrechts: the correlation matrix P is estimated from the standardized residuals $z_{i,t} = r_{i,t}/\sigma_{i,t}$, and the portfolio conditional variance is $\sigma^2_{p,t} = w'\Delta_t P \Delta_t w$, where Δ_t is the diagonal matrix of conditional volatilities. VaR and ES use the normal formula applied to the last estimated conditional volatility. All six GARCH models converged.

3.5 Filtered Historical Simulation with EWMA

The FHS method sits between historical simulation and GARCH. Instead of fitting a full GARCH model, we use EWMA filter with $\lambda = 0.94$, to track portfolio variance:

$$\hat{\sigma}_t^2 = \lambda \hat{\sigma}_{t-1}^2 + (1 - \lambda)\, r_{t-1}^{p\,2}.$$

We then standardize the returns by the EWMA volatility and take the empirical quantile of those residuals as the VaR input. This way we get time-varying volatility without having to use a parametric distribution for the residuals.

3.6 Univariate VaR and ES per asset

Before combining the assets into a portfolio, Table 2 reports one-day VaR and ES for each individual risk factor under three methods.

Table 2: Univariate one-day VaR and ES per asset at the 99% confidence level

Asset	$\mathrm{VaR_N}$ (%)	$\mathrm{ES_N}$ (%)	VaR_t (%)	ES_t (%)	$\mathrm{VaR_{HS}}$ (%)	$\mathrm{ES_{HS}}$ (%)
IBM	3.34	3.82	3.74	4.95	4.36	6.34
MSFT	3.67	4.22	4.12	5.49	4.17	6.21
MTOR	7.42	8.51	8.32	11.03	8.68	11.48
S&P 500	2.50	2.87	2.81	3.73	3.22	4.82
EUR/USD	1.20	1.38	1.35	1.78	1.41	1.71
Loan	*Near-zero by construction — floating rate instrument*					

Notes: $\mathrm{VaR_N}$ and $\mathrm{ES_N}$ assume normality. VaR_t and ES_t use Student-t with $\nu = 5$. $\mathrm{VaR_{HS}}$ and $\mathrm{ES_{HS}}$ are from historical simulation. The loan return is near-deterministic so VaR and ES are not meaningful for this component.

MTOR stands out as the riskiest individual asset, with a historical simulation VaR of 8.68% and ES of 11.48%, consistent with its high volatility shown in Table 1. The S&P 500 and EUR/USD carry significant lower individual risk. Across all risky assets, the Student-t and historical simulation estimates exceed the normal VaR, reflecting the fat tails already visible in the descriptive statistics.

3.7 Results

Table 3 collects the results. The unconditional methods, Normal, Student-t and historical simulation, all produce higher estimates, which is not surprising given that the March 2020 crash is sitting in the tail of the sample. The conditional methods tell us something else: by end of December 2020 volatility had come down significantly, with both GARCH and FHS estimating portfolio volatililty below 0.70%, pulling VaR and ES down with it. Both methods answer different questions: if the goal is long-run capital planning, unconditional estimates are more appropriate. For day-to-day risk monitoring, conditional methods that actually react to current market conditions make more sense.

Table 3: One-day VaR and ES estimates at the 99% confidence level

Method	VaR (%)	ES (%)
Normal Variance-Covariance	2.13	2.45
Student-t ($\nu = 5$)	2.40	3.18
Historical Simulation	2.67	3.76
GARCH(1,1) + CCC	1.46	1.67
Filtered Historical Simulation	1.74	2.18

Notes: 99% confidence level, one-day horizon, full sample 30 March 2011 – 31 December 2020. Conditional methods reflect market conditions as of 31 December 2020.

4 Backtesting

We evaluate how well the VaR estimates performed out-of-sample using a rolling 250-day window for the three unconditional methods. Each day we re-estimate VaR and ES using only the previous 250 observations, then check if the actual loss exceeded the prediction. For GARCH and FHS we do not use a separate rolling window because their volatility estimates are already time-varying. A violation is defined as:

$$I_{t+1} = \mathbf{1}_{\{L_{t+1} > \widehat{\mathrm{VaR}}_t\}},$$

and under a correctly specified 99% model the expected violation rate would be:

$$\hat{p} = \frac{1}{T} \sum_{t=1}^{T} I_{t+1} \approx 1 - \alpha = 0.01,$$

which is roughly 2.5 violations per year.

4.1 Violation counts

Table 4 shows the violations per year for all five methods. In calm years like 2013, 2016 and 2017 most methods stay close to the expected 2.5. The issues show up in 2018 and 2020. In 2018 the Normal method produced 14 violations, almost six times the expected number. Student-t 12, Historical Simulation 6 and GARCH and FHS at 6 and 5. None of the unconditional methods saw the sell-off in Q4 of 2018 coming. In 2020, the COVID crash breaks all our methods, with violations ranging from 8 to 13, which is much higher than the expected 2.5. There was nothing like the crisis of March 2020 in the 250-day window, so this is not surprising. It does, however, show the main limitation of risk models that look back in time. Figure 2 shows how the violations have changed over time. The Normal and Student-t VaR lines don't move much during the sample, they don't change when the market does. There is a clear difference between GARCH and FHS: their VaR lines go up sharply during the volatility surge in 2020. The problem is that the spike happens after the violations, not before.

Table 4: VaR violations per year across all five methods

Year	Expected	Normal	Student-t	Hist. Sim	GARCH	FHS
2012	1.9	0	0	0	2	0
2013	2.5	2	2	2	2	2
2014	2.5	6	3	5	2	5
2015	2.5	6	5	5	4	4
2016	2.5	2	2	2	3	2
2017	2.5	2	1	1	1	3
2018	2.5	14	12	6	6	5
2019	2.5	4	4	1	6	4
2020	2.5	13	12	8	8	8

Notes: Rolling 250-day window for Normal, Student-t and Historical Simulation. GARCH and FHS use time-varying conditional volatility estimates directly. Expected violations based on 99% confidence level.

4.2 ES backtest

Table 5 shows how the predicted ES compares to the actual average shortfall on days when there are violations. The Normal ES is wrong almost every year. It predicted 2.44% in 2016, but the actual shortfall was 3.25%. It predicted 2.64% in 2020, but the actual average loss on violation days was 4.06%. Student-t does a lot better: it hits the actual shortfall exactly in 2018 and comes much closer in most other years. It still underestimates in 2020, but the difference is smaller than for Normal. This fits with what the QQ-plots already said: fat tails are real, and the normal distribution can't handle them.

Table 5: Predicted ES versus actual shortfall on violation days

Year	Actual shortfall (%)	Predicted ES Normal (%)	Predicted ES Student-t (%)
2013	2.18	1.94	2.53
2014	1.65	1.63	2.22
2015	2.21	1.86	2.43
2016	3.25	2.44	3.16
2017	1.40	1.42	2.10
2018	2.13	1.63	2.13
2019	2.36	2.15	2.79
2020	4.06	2.64	3.31

Notes: Actual shortfall is the average portfolio loss on days where the Normal VaR was breached. 2012 excluded — no violations that year.

4.3 Dependence in violations

If we set up a VaR model correctly, violations should happen at random and not depend on each other.A breach today shouldn't tell us anything about what will happen tomorrow. Figure 3 shows that this is not the case for any of our methods. All three show a lot of autocorrelation at more than one lag, and this is higher than the 95% confidence bands. Violations happen in groups, not at random times. We see this directly in the violation plots for 2018 and 2020, where the red dots group together instead of spreading out. The most persistent clustering occurs in Student-t, while Historical Simulation exhibits the least persistent clustering. None of the models meet a basic test of independence.

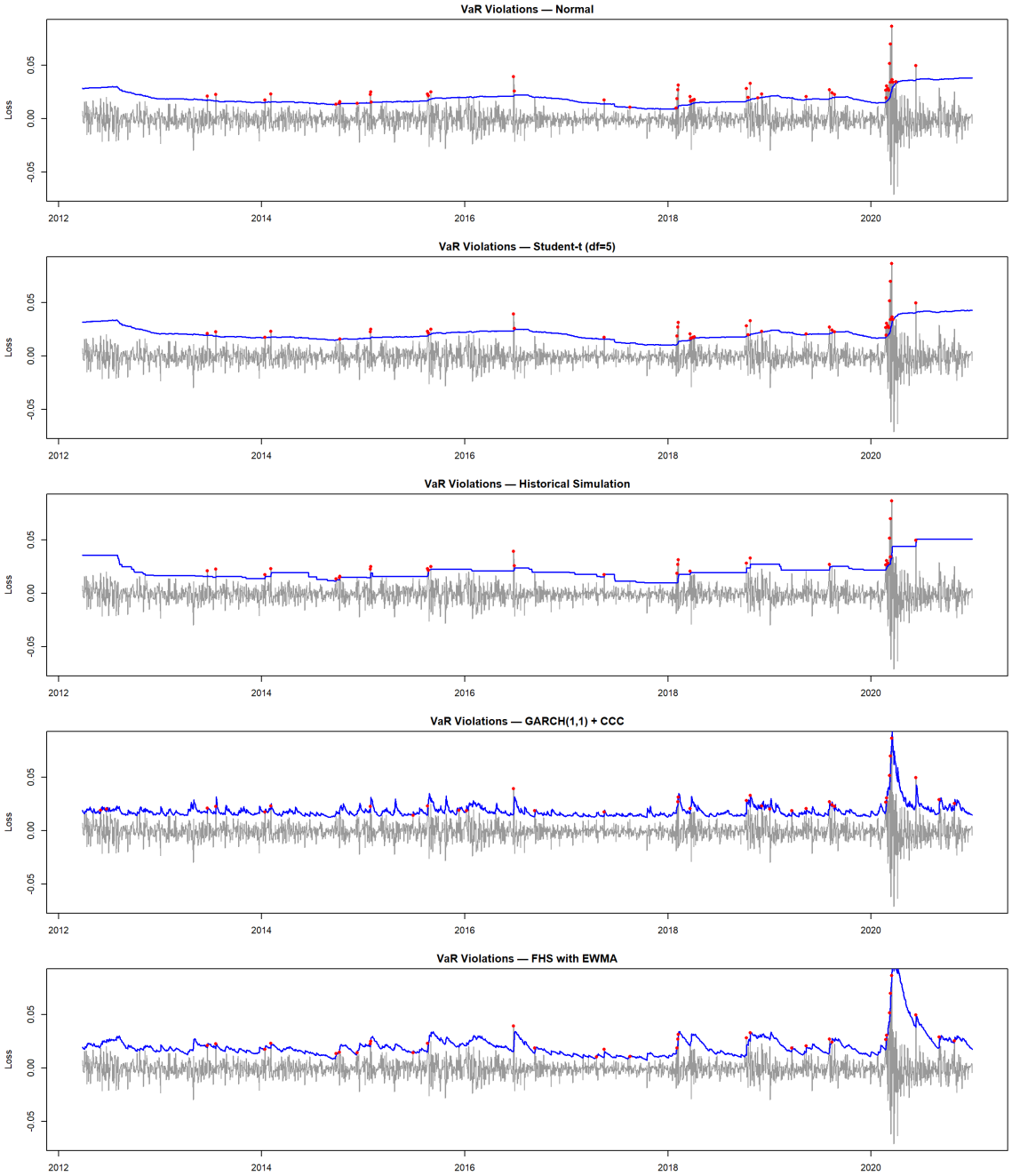

Figure 2: VaR violations for all five methods. Grey line is daily portfolio loss, blue line is rolling VaR estimate, red dots mark violation days.

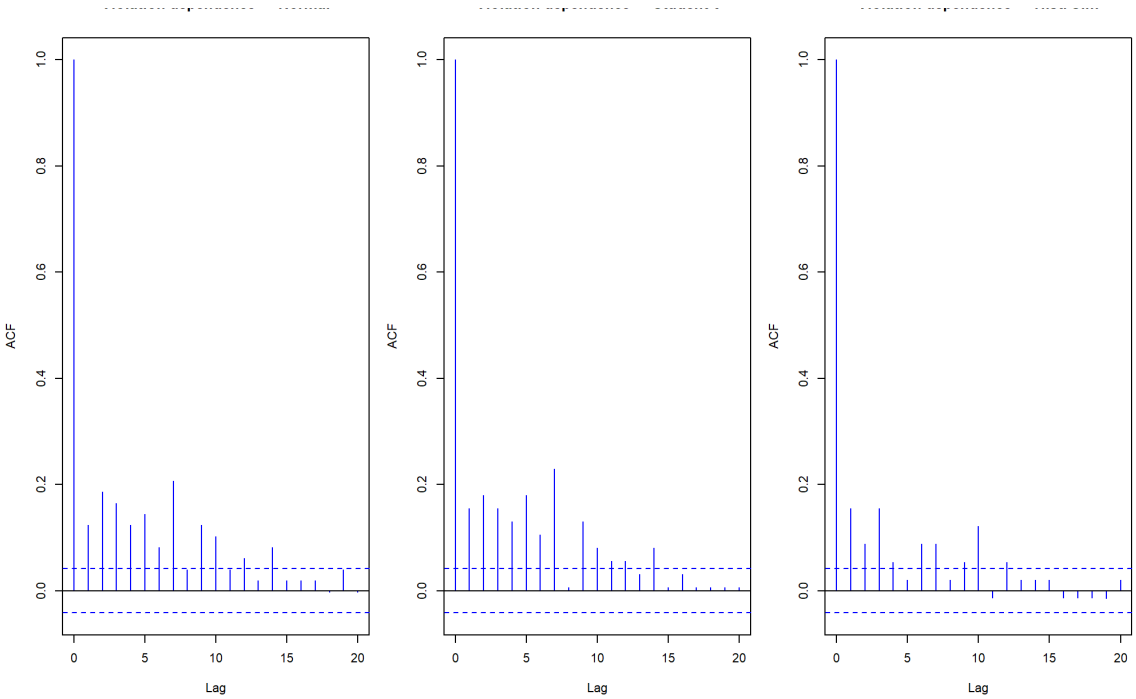

Figure 3: Autocorrelation function of violation indicators for Normal, Student-t and Historical Simulation. Dashed lines indicate 95% confidence bands.

5 Multi-day VaR

Up to this point, we have only looked at one-day VaR and ES. To examine how portfolio risk changes over longer holding periods, we extend the historical simulation approach to 5-day and 10-day horizons using non overlapping portfolio returns. Since the portfolio series is expressed in simple returns, the multi-day returns are constructed by compounding within each block:

$$r^{(h)} = \prod_{j=1}^{h}(1 + r_{t+j-1}) - 1,$$

for $h \in \{5, 10\}$. The corresponding multi-day losses are then defined as $L^{(h)} = -r^{(h)}$, and the 99% VaR is obtained from the empirical quantile of these non-overlapping losses.

These estimates are then compared with the square root of time approximation,

$$\mathrm{VaR}(h) \approx \sqrt{h}\,\mathrm{VaR}(1),$$

which assumes returns are uncorrelated across days.

Table 6: Historical multi-day VaR at the 99% confidence level versus the square-root-of-time approximation

Horizon	Historical Simulation VaR (%)	Square-root-of-time VaR (%)
1-day	2.67	2.67
5-day	5.27	5.98
10-day	7.04	8.46

Notes: Multi-day historical VaR is based on non-overlapping compounded portfolio returns. The square-root-of-time approximation scales the 1-day historical VaR by $\sqrt{h}$.

The comparison shows that the square root of time rule gives VaR estimates that are a little higher than those from historical simulation over longer time frames. This is true for both the

5-day and 10-day time periods, when the scaled VaR is still higher than the empirical estimate. So, the square-root rule appears to be slightly conservative for this portfolio. The differences aren't big enough to make the approximation wrong, so it can still be used as a rough guide for measuring risk over multi-day horizons.

6 Stress Testing

Backtesting looks at what happened in the past, but stress testing puts a specific shock on the portfolio and sees how much it would lose. We use the portfolio as of 31 December 2020, and we only change one category at a time, leaving everything else the same.

6.1 Setup

Portfolio loss under a shock is computed as:

$$L_{\text{stress}} = -\sum_i w_i \cdot \Delta r_i^{\text{shock}},$$

where w_i is the portfolio weight and $\Delta r_i^{\text{shock}}$ is the imposed shock return. Since the portfolio has no commodity exposure, that category does not apply. The relevant shock categories are equity, FX and interest rates.

The equity shocks hit IBM, MSFT, MTOR and the S&P 500 jointly, giving a combined equity weight of 70%. EUR/USD qualifies as a major currency so we apply $\pm 10\%$ and $\pm 20\%$ shocks. The interest rate shock works through the loan component: a shift of Δy changes the daily loan return by $\Delta y/252$, scaled by the 15% loan weight.

The math already tells us what to expect before we run anything. A -20% equity move with a 70% weight gives a 14% portfolio loss, which is well above any of our VaR estimates. The FX shock at 15% weight and -10% gives 1.5%, which is below VaR. Since the loan has a floating rate, interest rate shocks should not matter much, a 200bp shift gives about $0.02/252 \times 15\% \approx 0.001\%$ per day, essentially nothing.

6.2 Results

Table 7 reports the portfolio loss under each scenario.

Table 7: Stress test results: portfolio loss under instantaneous shocks

Scenario	Loss (%)	Ratio to VaR_N	$>\text{VaR}_\text{N}$	$>\text{VaR}_\text{HS}$
Equity -40%	28.00	13.15×	Yes	Yes
Equity -20%	14.00	6.57×	Yes	Yes
Equity $+20\%$	-14.00	—	No	No
Equity $+40\%$	-28.00	—	No	No
FX -20%	3.00	1.41×	Yes	Yes
FX -10%	1.50	0.70×	No	No
FX $+10\%$	-1.50	—	No	No
FX $+20\%$	-3.00	—	No	No
Rate +200bp	0.001	$< 0.01\times$	No	No
Rate -200bp	-0.001	—	No	No
Rate +300bp	0.002	$< 0.01\times$	No	No
Rate -300bp	-0.002	—	No	No

Notes: Shocks applied instantaneously as of 31 December 2020. VaR_N = Normal VaR (2.13%), VaR_HS = Historical Simulation VaR (2.67%). Gain scenarios not reported in ratio column.

6.3 Interpretation

Everything turned out as expected. Equity is the biggest risk by far. A move of -20% takes away 14% of the portfolio, which is more than six times the Normal VaR. At -40%, it hits 28%, which is about what happened on the worst days of COVID, but all in one day. This backs up what the backtesting already showed: VaR at 99% just can't handle shocks this big. The FX results are more interesting than expected. A 10% drop in EUR/USD means a 1.5% loss, which is less than our VaR estimates, so our models already take that into account. But a move of -20% gives 3%, which is more than the Normal VaR of 2.13% and close to the Historical Simulation VaR of 2.67%. This is something to keep in mind for a portfolio with 15% FX exposure. Interest rates don't really matter. A 200bp change means a daily loss of less than 0.001%. This is only because of how the loan is set up. A floating rate that changes every day has no effect on duration, but a fixed-rate bond would see its value change when rates change. The most important thing to remember is that equity risk is what drives risk in this portfolio. The 70% combined weight means that any big change in the market will cause losses that a standard VaR model at 99% can't handle.

7 Discussion and conclusion

7.1 Discussion

The central tension in this analysis is between two different questions a risk manager might ask: what is the long-run risk of this portfolio, and what is the risk right now? The unconditional and conditional methods essentially answer these two questions separately, and comparing them is more informative than asking which one is "correct."

Among the unconditional methods, the Normal assumption is the obvious weak point. The kurtosis figures in Table 1 already signal this before any model is estimated, and the backtesting results confirm it: 14 violations in 2018 alone, nearly six times the expected number. Switching to a Student-t with $\nu = 5$ is a genuine improvement, especially for ES, where it tracks the actual shortfall far more closely in stressed years. This matters in practice: ES underprediction directly implies insufficient capital buffers, and the Normal model fails on this dimension almost every year. Historical simulation sidesteps the distributional question entirely and ends up the most conservative of the three, largely because the COVID crash is sitting in the empirical tail, a feature of the sample rather than a property of the method.

The conditional methods, GARCH and FHS, are doing something fundamentally different. By December 2020, after volatility had settled, both produce VaR estimates well below the unconditional benchmarks. This is not a failure; it reflects genuinely lower short-term risk at that point in time. The practical implication is straightforward: conditional models belong in daily risk monitoring, unconditional models belong in capital planning.

What none of the methods solve, however, is violation clustering. All three unconditional models show strongly autocorrelated violations in 2018 and 2020, well beyond the 95% confidence bands. Even the conditional models, which react faster, produce clustered violations because the volatility spike always arrives before the model can incorporate it. This is the deeper limitation of the exercise: the problem is not which distribution to assume, but that all backward-looking models are structurally slow during regime shifts. A risk analyst relying solely on any of these methods during a crisis would consistently underestimate risk precisely when it matters most.

The stress tests, discussed in detail in Section 6, reinforce this conclusion. VaR is a normal-times risk measure, and for this portfolio with 70% combined equity exposure, a genuine market crisis will always produce losses that fall well outside what any 99% confidence level can capture.

7.2 Limitations

Several limitations of this analysis are worth acknowledging. The most immediate is the use of fixed portfolio weights throughout the entire sample period. While this simplifies the risk attribution and makes VaR and ES estimates easier to interpret, it is not realistic over a ten-year horizon. A portfolio manager would almost certainly rebalance, and the resulting return series would look quite different.

The loan component is a second simplification. Modelling it as a floating-rate proxy tied to the 3-month Treasury bill rate plus a fixed spread captures the broad behaviour of a short-term credit instrument, but it is not a fully priced credit security. As a result, the interest rate risk in the portfolio is likely understated, and the stress test results for rate shocks should be interpreted with that in mind.

A subtler but equally important limitation is the violation clustering documented in Section 4. None of the five methods produce independent violations: breaches cluster in 2018 and 2020, well beyond what a correctly specified model should generate. This is not a problem that can be fixed by choosing a fatter-tailed distribution. It reflects the structural inability of backward-looking models to respond quickly enough during regime shifts. A dynamic model that explicitly accounts for volatility regime changes, such as a Markov-switching specification, would likely perform better on this dimension.

Finally, the sample ends in December 2020, which means the 2022 rate hiking cycle and subsequent market dislocations are not captured. A longer sample would likely raise the unconditional VaR estimates and provide a more demanding backtesting environment for all five methods.

7.3 Conclusion

This analysis set out to measure and predict portfolio risk using five methods that differ in their distributional assumptions and treatment of volatility. The results are consistent enough to draw some clear conclusions, but also honest about what none of the methods can do.

For a risk analyst choosing between these five approaches, the recommendation depends on the purpose. For day-to-day risk monitoring, the conditional methods are the right tool: GARCH and FHS respond to current market conditions and produce estimates that reflect where volatility actually is, rather than where it was on average over the past decade. For capital planning and regulatory purposes, the Student-t with five degrees of freedom is the most defensible unconditional method. It captures fat tails better than the Normal and tracks actual shortfall more closely in stressed periods, as the ES backtest makes clear. The Normal variance-covariance method should not be used as a standalone risk measure for this portfolio; the backtesting results disqualify it.

What all five methods share, however, is a structural weakness during crises. Violation clustering in 2018 and 2020 shows that no backward-looking model adapts quickly enough when the market shifts regime. This is not a reason to abandon VaR and ES as measures, but it is a reason to supplement them. Stress testing fills part of this gap by making explicit what a 99% VaR cannot show: that a severe but plausible equity shock can produce losses many times larger than any daily VaR estimate. For this portfolio, with 70% combined equity exposure, that is the dominant risk and it deserves more attention than a single confidence level can provide.

References

[1] Bos, C.S. (2025/2026). *Quantitative Financial Risk Management*. Lecture notes, Duisenberg Honours Programme in Quantitative Risk Management, Vrije Universiteit Amsterdam.

[2] McNeil, A.J., Frey, R., and Embrechts, P. (2015). *Quantitative Risk Management: Concepts, Techniques, and Tools*. Revised edition. Princeton University Press, Princeton.

[3] Center for Research in Security Prices (CRSP) (2026). *Daily total return data for individual stocks and the S&P 500 index*. CRSP, University of Chicago Booth School of Business. `https://www.crsp.org`

[4] Board of Governors of the Federal Reserve System (US) (2026). *3-Month Treasury Bill Secondary Market Rate, Discount Basis [DTB3]*. Retrieved from FRED, Federal Reserve Bank of St. Louis. `https://fred.stlouisfed.org/series/DTB3`

[5] Board of Governors of the Federal Reserve System (US) (2026). *U.S. Dollars to Euro Spot Exchange Rate [DEXUSEU]*. Retrieved from FRED, Federal Reserve Bank of St. Louis. `https://fred.stlouisfed.org/series/DEXUSEU`